
Promoting sustainable data quality management to enhance HIV programmes

Country-level implementation, lessons learned and future directions, including embracing artificial intelligence

Technical brief



World Health
Organization

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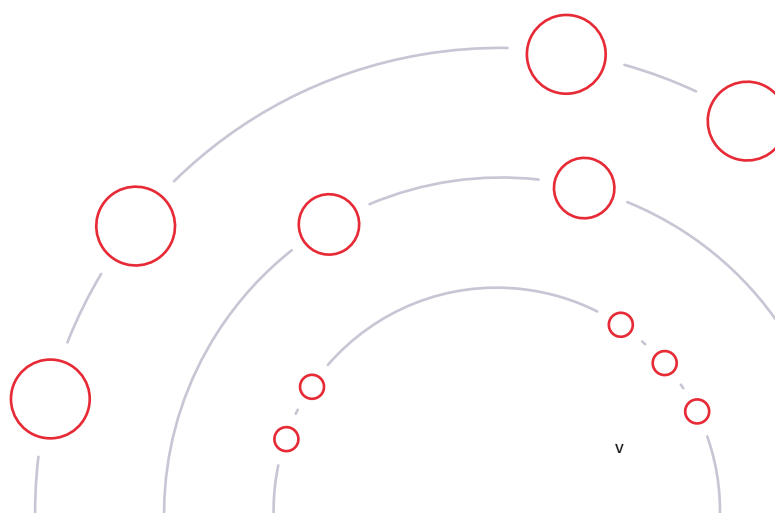
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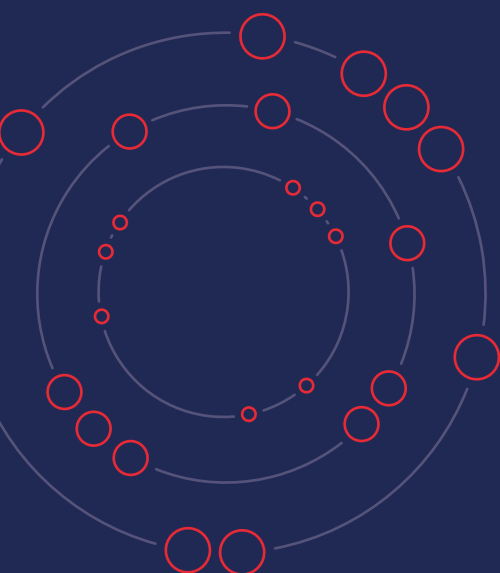
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Abbreviations and acronyms

AI	artificial intelligence
ART	antiretroviral therapy
DHIS2	District Health Information Software 2
DQA	data quality assessment
DQI	data quality improvement
Global Fund	Global Fund to Fight AIDS, Tuberculosis and Malaria
PCR	polymerase chain reaction
PEPFAR	United States President's Emergency Plan for AIDS Relief
UNAIDS	Joint United Nations Programme on HIV/AIDS
UNICEF	United Nations Children's Fund



1. Background: the data quality imperative

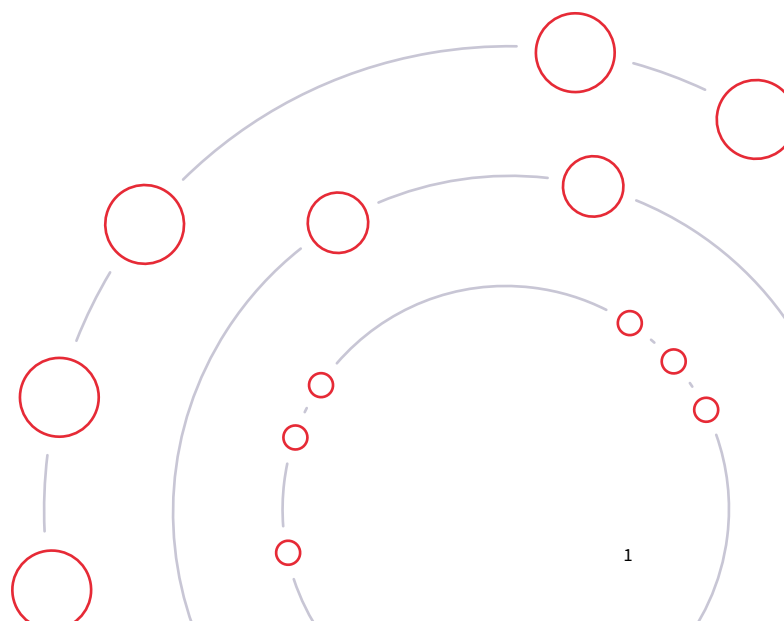


High-quality data are essential for effective HIV programmes, enabling efficient service delivery, resource allocation and monitoring of progress, accurate budgeting and evidence-informed decision-making. Improved data quality and utilization, especially at a granular level, can enhance programme impact, focus the provision of efforts for specific populations and in geographical areas and advance a sustainable HIV response towards universal health coverage.

The WHO 2022 person-centred HIV strategic information guidelines (1) emphasize the importance of countries developing strategies to improve data quality and use, integrating data quality assessment (DQA) with corrective actions to improve country data systems. To facilitate this, collaborative DQA tools for verifying HIV treatment, viral load testing data and patient monitoring systems have been jointly developed by WHO, the Joint United Nations Programme on HIV/AIDS (UNAIDS), the United States President's Emergency Plan for AIDS Relief (PEPFAR) and the Global Fund to Fight AIDS, Tuberculosis and Malaria (2,3). In 2024, the second editions of these tools were published (4,5) to further assist countries in improving their data quality, implementing corrective measures and strengthening data systems. Although the tools focus on HIV treatment and viral load testing data, countries are urged to incorporate key programmatic indicators for comprehensive assessments. Prioritizing future development of tools and approaches for data quality assurance for HIV testing services going forward can ensure that accurate and timely data are available to effectively target testing services, improve the monitoring of outcomes throughout the entire HIV care cascade and strengthen overall data systems.

Data quality improvement (DQI) involves a wide range of activities beyond DQAs tailored to each country's unique context, including data collection systems, monitoring and evaluation processes, available resources and existing quality improvement mechanisms. Integrating these processes throughout the data management cycle may ensure sustained DQI, and countries are encouraged to transition from focusing on DQA to the goal of data quality management, including DQI. However, the true value of DQI lies in leveraging improved data utilization to enhance service delivery and evidence-based decision-making through increased confidence in data quality, ultimately improving clinical care, health-care services and ensuring a sustainable HIV response.

This technical brief highlights country experiences, lessons learned and future directions in data quality management, including improvement for HIV programmes, aiming to share successful practices and underscore the pivotal role of DQI in driving programme enhancement and efficiencies and improving overall data quality. It also provides technical information on digital tools and artificial intelligence (AI) and their use in improving data quality.

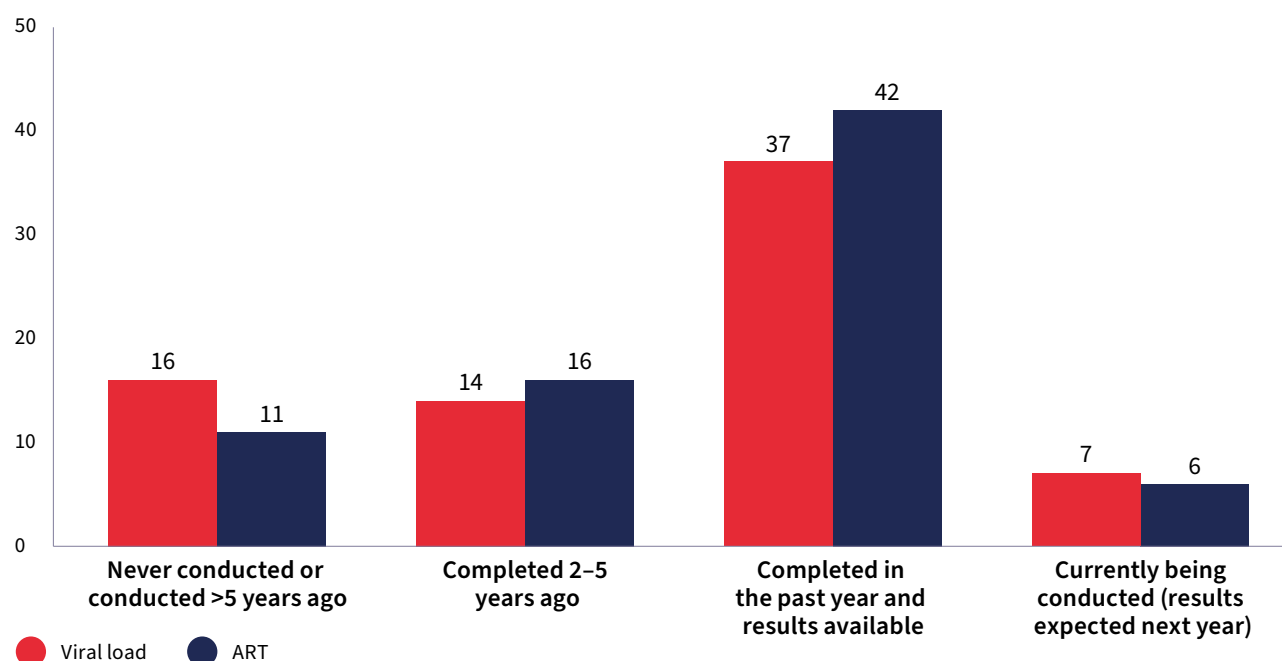


2. Country implementation of DQA

Countries and partners have made concerted efforts and substantial investments in improving data quality by implementing DQA as well as DQI activities more broadly. A WHO-convened coordinating mechanism was established between global partners to plan country technical assistance, coordinate financial support, monitor implementation, facilitate follow-up remedial actions to strengthen HIV patient monitoring systems and adjust national HIV treatment and viral load suppression estimates as needed. This provides a common platform for corrective actions, to improve patient monitoring and to learn from approaches across partners and countries.

In 2024, 42 countries reported implementing DQA of HIV treatment data, with the results available via the Global AIDS Monitoring. In addition, 37 countries reported conducting a DQA for viral load suppression data, with about 90% using the results to adjust their national estimates of the number of people receiving antiretroviral therapy (ART) and with suppressed viral loads. The highest number of countries reporting conducting DQAs for both HIV treatment and viral load testing data was in sub-Saharan Africa, Latin America and the Caribbean (Fig. 1). A further six countries planned or were already implementing DQAs for ART, and seven countries had similar plans for viral load testing data, demonstrating continued prioritization and investment in DQA activities by countries and partners.

Fig. 1. Countries reporting implementation of DQA of HIV treatment and viral load testing data in 2024, global



Source: 2024 data from Global AIDS Monitoring (UNAIDS/WHO/UNICEF).

3. Programmatic experience and lessons learnt

This section summarizes the strategies and approaches that have supported successful implementation of DQI activities, including DQAs, along with valuable insights gained from country experience (Fig. 2). These

aim to enable countries to maximize the investments and benefits of these exercises in improving data, strengthening data systems, driving programme improvements and enhancing data use.

Fig 2. Lessons learned on improving data quality within HIV programmes from country implementation experience



01 Importance of multistakeholder engagement



02 Ensure follow-up of DQAs and implementation of DQI activities



03 Link DQI with programme improvement



04 Role of DQI in strengthening investment impact



05 Use of DQA results

3.1 Importance of multistakeholder engagement

One fundamental lesson learned is the importance of engaging a range of stakeholders in DQI, including DQA exercises, to routinize these activities as integral components of effective HIV programmes to build sustainability. Securing full ownership and support for data quality activities from ministries of health ministries and partners is an important part of this lesson. Successful implementation can be promoted through Ministry of health leadership and oversight of the overall process and implementation of remedial actions with stakeholders in clearly defined roles, such as providing funding and technical assistance, overseeing implementing partners and ensuring the use of harmonized tools and collaborating with country teams to adjust national data and estimates. Establishing an interagency country team comprising key stakeholders, including civil society participation, is another important lesson learned for collaborative implementation, follow-up of remedial actions and ensuring accountability. This enables corrective actions to be distributed among partners to improve data and data systems.

3.2. Ensuring follow-up of DQA and implementation of DQI activities

DQA implementation involves developing an action plan to address identified data quality challenges and facilitate remedial actions for resolution, including a follow-up mechanism to monitor data quality issues and the implementation of DQI activities for which tools are available. This can include developing and using a database to track data quality issues and the implementation of remedial actions (4).

Countries have undertaken various DQI activities based on DQA findings, ranging from building the capacity of health-care workers and personnel involved in data reporting and management, updating reporting tools and electronic medical records systems, client verification initiatives and data deduplication (5). These activities have helped to address the underlying issues contributing to data quality challenges and have strengthened national data systems. Box 1 provides an example of DQI activities carried out following DQA in Ethiopia.

Box 1. DQI activities implemented following DQA in Ethiopia

For the past six years, the United States Centers for Disease Control and Prevention in collaboration with Ethiopia's Federal Ministry of Health has promoted intensive data quality assurance as well as DQI activities to advance HIV and TB epidemic control to achieve the 95–95–95 targets for HIV testing and treatment.¹ Ethiopia's programme for eliminating mother-to-child transmission of HIV identified programmatic gaps and data discrepancies in the SPECTRUM HIV modelling; a suspected overestimation of children living with HIV identifying an opportunity to enhance the quality of data monitoring of the elimination of mother-to-child transmission and maternal, newborn and child health programme.

DQA, including completeness, validity, consistency and verification with District Health Information Software 2 (DHIS2) of the Ethiopia's Federal Ministry of Health was conducted between July 2022 and June 2023. Data were collected from 139 high-volume health-care facilities from the 10 regions and two city administrations in Ethiopia, which represents 5% of the facilities providing services for eliminating mother-to-child transmission of HIV in Ethiopia. Quantitative and qualitative assessment captured the effectiveness of data management systems, data use for programme improvement, and workforce capacity to sustain routine data quality monitoring.

Twelve data elements were assessed from 2420 unique records of mothers living with HIV and 2191 HIV-exposed infants from the mother-baby cohort registers. In addition, nine reportable indicators were examined for data verification between electronic medical records and DHIS2 as well as PEPFAR reporting systems. Of the 12 data elements reviewed for record completeness, both the maternal and HIV-exposed infant charts showed higher percentages of completed and valid data than the respective registers. Data completeness was higher in patient charts (92%) than registers (87%), with syphilis testing (69.3%) and viral load testing (52.8%) data being the least complete (Table 1). Data consistency between patient charts and registers was less than 90%, with isoniazid start date and infant enrolment having the lowest consistency (Table 1). The register for the elimination of mother-to-child transmission of HIV has a higher completeness percentage than the paper-based patient charts for the data elements related to laboratory test results: syphilis test results and DNA polymerase chain reaction (PCR) results for early infant diagnosis. The DQA team found that some of the laboratory results that were paper forms attached to the patient charts were lost in due course of time. Since laboratory results were attached with the patient charts but not transcribed in the patient history chart, more documentation is present in the register for the elimination of mother-to-child transmission of HIV than the patient chart. Hence one of the recommendations in the DQA is to transcribe laboratory results in the patient history chart electronically in addition to attaching the laboratory paper result to address such inadvertent data incompleteness issues.

¹ 95% of people living with HIV know their HIV status; 95% of people living with HIV who know their HIV status are receiving treatment; 95% of people living with HIV receiving treatment have suppressed viral loads.

Box 1. (continued) DQI activities implemented following DQA in Ethiopia

Data verification showed underreporting of pregnant women receiving ART (15%) and HIV-exposed infants receiving co-trimoxazole (21%) compared with the national reporting system (DHIS2).

The major data quality issues identified were underreporting and inconsistency, potentially leading to inaccurate estimation of mother-to-child transmission rates of HIV and the number of children living with HIV. The DQA intervention on the elimination of mother-to-child transmission led by Ethiopia's Federal Ministry of Health and regional health bureaus integrated training not only on DQA standards but also on data definitions to enhance the quality (completeness) of data entry into mother-baby cohort registers, promoting the use of data on the elimination of mother-to-child transmission of HIV for programmatic actions and routinizing ongoing technical support. This DQA led to the development and implementation of a module on the elimination of mother-to-child transmission of HIV for Ethiopia's electronic medical records system, which is anticipated to support DQI and data use. The validated data captured from the DQA of elimination of mother-to-child transmission of HIV led to a reduction in the estimates of the number of children living with HIV by 16% compared with 2023 projections based on modelling (from 33 026 to 27 720). Since only 5% of the health-care facilities for the elimination of mother-to-child transmission of HIV were included in the initiative initially because of resource constraints, this impactful effort has led to a commitment by Ethiopia's Federal Ministry of Health and regional health bureaus to expanding and routinizing the DQA on the elimination of mother-to-child transmission of HIV to additional sites and scaling up these DQI efforts throughout the country.

Table 1. Data completeness comparison of 12 data elements between patient charts and ART registers for the elimination of mother-to-child transmission, Ethiopia, July 2022–June 2023

Data element completeness		Patient chart	Register for the elimination of mother-to-child transmission
Elimination of mother to child transmission of HIV			
1	Age (n = 2420)	99.9%	99.7%
2	Date of entry into the register for elimination of mother-to-child transmission (n = 2417)	96.2%	99.2%
3	Syphilis test result (n = 2306)	69.3%	81.4%
4	Date ART initiated (n = 2318)	97.9%	94.1%
5	ART regimen at last visit (n = 2319)	99.1%	98.0%
6	TB screening visit (n = 2318)	97.3%	88.5%
7	TB preventive treatment start date (n = 2393)	89.5%	75.5%
8	Latest viral load test result visit (n = 2308)	71.8%	52.8%
Early infant diagnosis			
9	Date of infant enrolment (n = 2191)	89.1%	90.6%
10	Date infant received antiretroviral prophylaxis visit (n = 2185)	90.2%	89.1%
11	Result of DNA PCR for infant visit (n = 2085)	80.0%	86.1%
12	Co-trimoxazole treatment started visit (n = 2151)	90.4%	91.9%

Source: Ethiopia's Federal Ministry of Health, 2024

3.3. Linking DQI with programme improvement

Programme improvement is a continuous endeavour aimed at achieving measurable changes in programme performance through strategic data-driven changes to enhance programme design, implementation, quality and impact. It relies on high-quality data to inform adjustments that optimize programmes and address service delivery gaps. The approach is effectively implemented through a series of time-bound activity cycles involving multiple stakeholders emphasizing data triangulation, communication and rapid information to address specific programme and service issues (6). DQA and DQI initiatives have significantly contributed to improving programme quality across contexts.

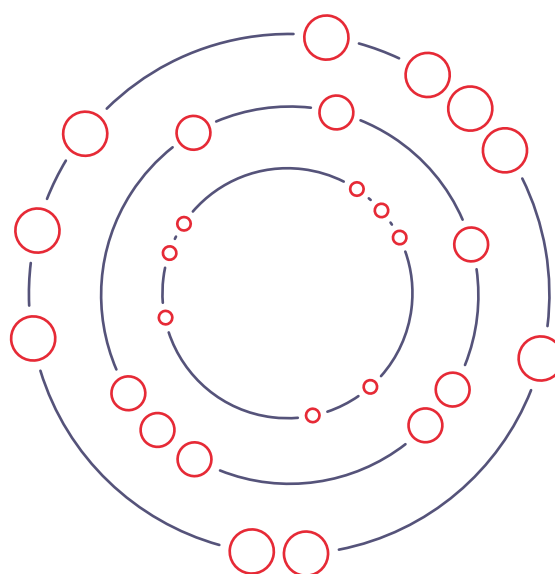
3.4. Role of DQI in strengthening investment impact

High-quality data may result in optimal allocation of resources and enable the design of interventions and programmes for maximum impact. Conversely, poor-quality data may divert resources from acute needs and create unnecessarily waste. Investing in data quality yields numerous benefits, including enhancing investment impact, delivering value for money by improving programme efficiency and maximizing programmatic impact through data-driven priority-setting of investment. Programme interventions based on high-quality targeted data provide an opportunity to support quality improvement and service delivery efficiency. For instance, accurate and timely data on the number of people receiving ART and newly initiating ART informs antiretroviral drug forecasting and procurement and efficient supply chain management.

3.5. Using DQA results

In 2024, most countries (38 of 42) reporting to the Global AIDS Monitoring having implemented a DQA for HIV treatment data used the results to adjust their national estimates of treatment coverage. Similarly, 33 of the 37 countries reporting conducting DQA for viral load testing data in 2024 adjusted their national estimates of viral suppression using these results. This underscores the importance of DQA exercises in improving national data and HIV estimates, which feed into the annual HIV estimation process conducted annually by many countries to inform HIV programme planning and monitoring. In addition, timely DQA results are critical for informing the quantification and forecasting of HIV-related commodities, including antiretroviral drugs and viral load reagents.

“Investing in data quality yields numerous benefits, including enhancing investment impact, delivering value for money by improving programme efficiency and maximizing programmatic impact through data-driven priority-setting of investment”



4. Future directions for data quality management for HIV programmes

4.1 Transitioning to routine data quality monitoring and management

Significant efforts have been invested in implementing large-scale, typically nationally representative DQAs. These exercises demand substantial human and financial resources from countries and partners since they involve site level verification and recreation of indicators that are reported to the ministry of health and partners. Going forward amid concerns of the sustainability of funding for the HIV response, there is greater recognition that efforts to ensure data quality should not be limited to large-scale infrequent exercises but rather routine data quality management activities that are integrated into existing efforts to strengthen health information systems and long-term DQI strategies.

Data quality management is a part of sustainability. Sustainability in turn necessitates transitioning to alternative approaches that build on existing data quality management processes and rationalizing the use of existing effective approaches such as DQAs. To support countries with this transition, the following conditions or circumstances are suggested for implementing a national DQA:

- when **data quality issues are detected** through routine data quality monitoring activities or programme evaluation requiring further investigation and verification;
- when **significant data discrepancies exist** between the ministry of health and partner country programme data and key indicators;
- when **significant changes in reporting patterns of indicators** occur across the country during periods of disruptions caused by natural disasters, conflict, pandemics or social or political unrest and
- **periodically**, to ensure that accurate, timely and reliable data are available for programme improvements and monitoring progress, countries should consider conducting nationally representative DQAs every two to three years.

Countries can consider giving priority to and scaling up routine data quality monitoring and management activities as components of sustainability plans for HIV programmes. The WHO 2020 module for assessing and strengthening the quality of viral load testing and HIV patient monitoring systems (3) and its 2024 update (5) focus on routine data quality monitoring activities that are recommended for viral load testing data and can be implemented more frequently than national DQAs. These include data quality monitoring using lot-quality assurance sampling to compare the consistency and completeness of data elements across different data sources, monthly data quality reviews of key HIV care cascade data at health-care facilities and supportive supervision visits incorporating data consistency and completeness checks. These routine activities for monitoring data quality can be extended to other programmatic areas.

4.2 Leveraging digital tools and AI to improve data quality

Various digital tools have been deployed across different settings to improve the quality of data within HIV programmes. These include digital data collection tools, automated data validation checks for data cleaning, using AI and machine learning and data deduplication among others. Digital adaptation kits for HIV programmes are available to support countries in adapting WHO HIV clinical and data recommendations within digital systems, including standardizing data elements to improve data quality and use (7).

AI pertains to algorithms embedded within systems and tools, enabling them to autonomously learn from data and execute tasks without requiring explicit programming for each step by a human (8,9). AI offers significant benefits in improving person-centred strategic information and the quality of services. However, collecting health and HIV data raises privacy concerns, especially for vulnerable groups such as key populations. In contexts with limited legal protections, AI processing of sensitive data poses risks, such as breaches and reidentification. Safeguarding privacy is crucial when developing AI health tools,

and robust data protection measures should be put in place when these are deployed. Ensuring responsible and ethical AI usage in HIV programmes necessitates collaborative efforts from stakeholders such as policy-makers, programme managers, developers and civil society to mitigate emerging risks while leveraging the benefits for programme improvement and service delivery (10,11).

Box 2 highlights an example from Malawi demonstrating digital data collection, validation and real-time reporting of HIV testing services data using AI.

Box 2. Real time digital data collection for HIV testing services using AI in Malawi

The Directorate of HIV, STI and Viral Hepatitis in Malawi has spearheaded the first nationwide deployment of scannable paper registers for integrated testing services in sub-Saharan Africa. The tools capture facility and community-based rapid testing for HIV, syphilis and hepatitis B and HIV-self test distribution. Paper records are digitized with minimal effort and within minutes from a simple image taken with a low-cost smartphone at the facility or in the community. After uploading the image to a cloud-based server, AI-based algorithms are used to recognize handwriting and convert it into complete client-level electronic data. The AI algorithm was specifically trained on Malawian handwriting and performs with >99.9% accuracy. The high-quality granular electronic data are then used for generating automated weekly and monthly summaries that are sent to the smartphones at all facilities and can be accessed by programme staff at the district and national levels via a web-based data portal. Algorithmic validation is used to identify testing protocol breaches, inconsistencies and documentation errors. Automated DQA reports are immediately sent to the facilities for review and corrective action.

This efficient quality improvement system has been extremely useful for Malawi's highly decentralized testing programme, which has more than 2000 service delivery points and is otherwise very challenging to supervise. Introduction of this innovative technology played a crucial role in maintaining high-quality services during the transition to the three-test algorithm for HIV diagnosis, which added complexity for lay providers. Between November 2022 and June 2024, more than 6.4 million client records were successfully digitized and utilized for routine monitoring and evaluation and continuous quality improvement. A field evaluation report concluded this innovative digital solution facilitates efficient analysis and use of large volumes of routinely collected individual-level data, provides real-time data, reduces transcription errors, saves time and human resources for data entry and overall improves the quality of HIV testing services data and programme monitoring.

Source: Directorate of HIV, STI and Viral Hepatitis, Malawi, 2024.

4.2.1 Automated data quality assurance

Automated data quality assurance approaches that harness machine learning have been developed to detect anomalies in reported data. The United States Agency for International Development initially deployed these approaches in countries during the coronavirus disease 2019 (COVID-19) pandemic when in-person DQAs could not be implemented. They enable the rapid conduct of remote reviews of the quality of aggregate facility-level data with high accuracy and enable targeted facility visits and interventions to support data quality. This reduces costs, minimizes disruption to clinical services and saves time associated with these activities, enabling more frequent and routine implementation.

4.2.2 Client verification and authentication

Identity verification in health-care services helps to ensure accurate client tracking, appropriate care and data quality (12). HIV programmes have employed various approaches to ensure that individuals are accurately and repeatedly identified when they access services, including using identification cards, biometrics and unique identifiers. WHO currently recommends using anonymous unique health identifiers to help individuals identify themselves when accessing health-care services across different facilities and over time, thereby supporting person-centred care (1). Anonymous unique identifiers are beneficial due to their ease of use, low cost, and scalability, making them suitable for expanding across ART sites. They also enhance client acceptance by protecting privacy and improve data security by reducing the amount of personal information shared across systems.

Biometrics, which include fingerprints, facial recognition and iris scans, serve the dual functions of client identification to ensure uniqueness and avoid duplication and authentication to confirm that the same person is interacting with the system at each point of contact (13,14). Biometrics have been used in HIV programmes in certain settings, offering simplicity for users, speed, enhanced data quality and continuity of care through streamlined registration processes and data linkage across data systems and service points. However, similar outcomes can be achieved using unique identifiers that separate personally identifying information from health data.

Despite their benefits, biometric systems pose inherent risks, including the potential exclusion of vulnerable groups, such as key populations because of low acceptability stemming from concerns of stigma, discrimination or criminalization (15). Biases in algorithms also pose risks of discrimination based on age (with limited reliability for very young children), gender and gender identity or skin colour (16–18). Furthermore, collecting, storing and using biometric data present data protection and privacy risks and have ethical and human rights implications (19). Robust governance frameworks for biometric systems, including strong privacy provisions and complaint procedures, may mitigate these risks. Technical measures such as data encryption and separation enhance security (20), while comprehensive legal frameworks can safeguard individuals' privacy and data rights (21). For instance, the European Union's General Data Protection Regulation restricts the collection and processing of biometric data, necessitating strong safeguards and broader regulations on data protection and security for personal data (22). However, despite the importance of these measures, they do not guarantee client acceptability, especially for key populations in settings where they face criminalization. Actively involving affected populations in planning, designing, monitoring and evaluating these protection measures is one approach to maximize client acceptability.

4.2.3 Data deduplication

Deduplication of patient records is a process that involves applying deduplication procedures to identify duplicate data and then integrating these data into a unified client record. Deduplication of patient records serves to improve both data quality and patient management and clinical outcomes if used as a routine DQI activity for all digital data systems.

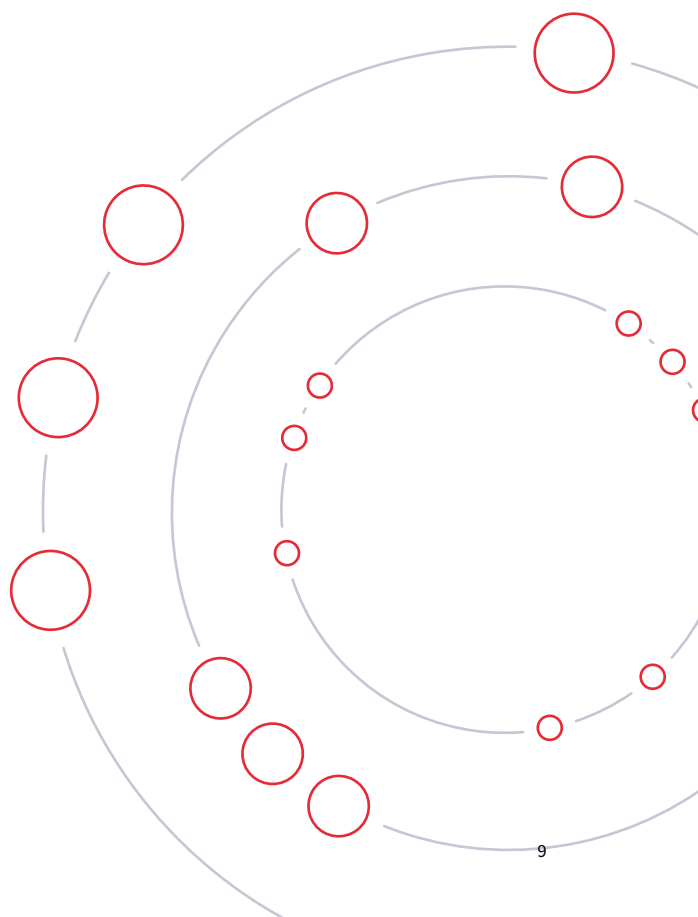
Interoperability for deduplication enables linking of distinct record systems for unification of client records. For instance, if duplicate records are correctly identified and merged, viral load testing data from the laboratory information management system and ART dispensing data from the pharmacy system can be linked and integrated into one comprehensive client record. Similarly, if community information systems are included into the process, data on service delivery at the community level can be exchanged with information in electronic medical records. This improves patient monitoring and care and is

particularly important with the scale-up of differentiated service delivery, in which HIV services may be provided in a variety of settings, including community settings.

Duplicate records can be created for a variety of different reasons. Clients who receive care at multiple sites or silently transfer to another health-care facility without documentation may have multiple client records, leading to overreporting of the number of people currently or newly initiating ART and being incorrectly classified as lost to follow-up. A client might also be registered multiple times with the same or different service delivery points, leading to the creation of multiple patient records within the same facility. As a result, health-care providers do not have insight into the full services received or client history, compromising integrated patient care. Fragmented reporting systems and poor architecture of data systems can also contribute to record duplication.

Deduplication entails identifying probable links in different data sets and verifying whether the probable link is a true link and using true links to link data from different data sets related to the same entity. Different methods can be used for deduplication, including deterministic or probabilistic linking methods, which compare the similarity of variables within data sets (23).

Box 3 highlights how deduplication has been successfully applied in Nigeria to improve data quality and service delivery.



Box 3. Deduplication of client records to support client verification, improve care and data quality in Nigeria

Recognizing that Nigeria continues to record active case-finding of new cases and approaching HIV treatment saturation for subpopulations and geographies, the programme commenced active client record deduplication. First, the country leveraged its high coverage of fingerprint capture for clients who are receiving treatment and the storage of encrypted templates to identify records with duplicate templates within a treatment facility or subnational unit or across subnational units. Subsequently, the programme utilized the templates to identify duplicate records for follow-up action. In addition to using fingerprint templates, the programme used deidentified demographic and clinical variables sourced from the client-level National Data Repository to match and identify potential duplicates. This process involved applying predefined criteria, such as multiple client medical records with same date of birth, sex and ART start date. In addition, records receiving treatment within the same facility but exhibiting matching ART start dates and at least three consecutive last antiretroviral drug pick-up dates were flagged for further examination (Fig. 3).

Records identified as potential duplicates underwent rigorous verification through client contact tracking, validation and resolution procedures. For duplicate records from different facilities that were found to belong to the same client, site-level staff engaged clients to determine their preferred treatment site, facilitating the consolidation of services and terminating duplicate records at affected sites. Duplicate records belonging to the same client in the same facility were merged into one. This approach improved the reliability of the data management systems to be able to uniquely identify clients who are receiving treatment across the supported treatment sites, enhance client verification mechanisms and ultimately bolster the quality of HIV care nationwide.

This process of client verification and deduplication has been further incorporated into routine data quality processes. The electronic medical records are equipped with functionalities to identify client records that may potentially be duplicated using fingerprint templates during enrolment and using the predefined criteria mentioned above to flag records for further verification. The National Data Repository continues to support the routine analysis and generation of records based on these criteria.

Source: National AIDS and STI Programme, Federal Ministry of Health, Nigeria, 2024.

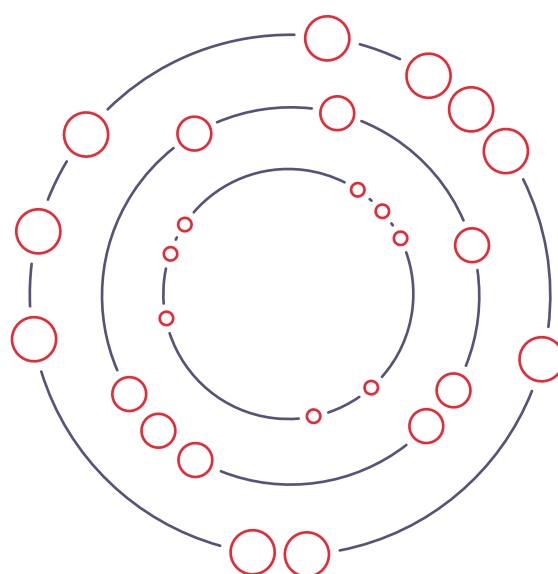
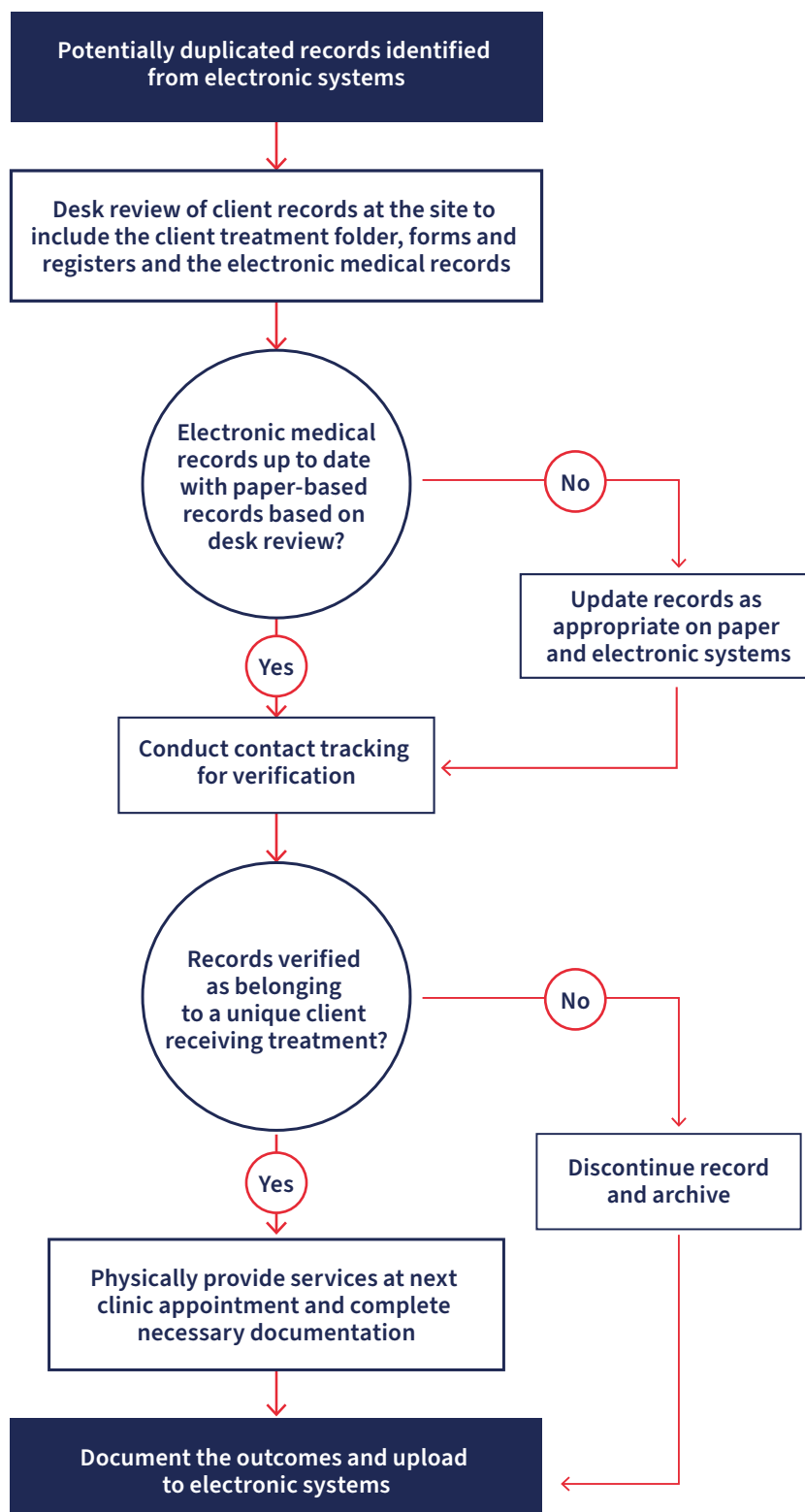


Fig. 3. Process for deduplicating client records to support verification and improve HIV care and data quality in Nigeria



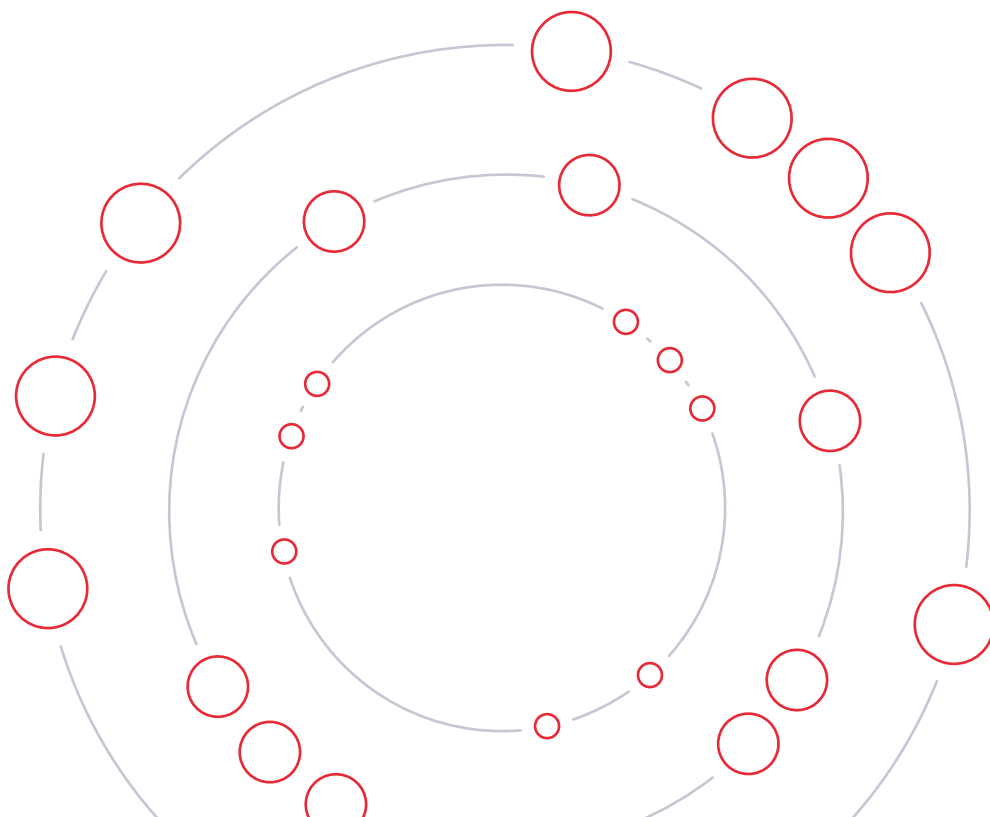
4.2.4 Client-level repositories

Client-level repositories serve as centralized clinical data registries that gather client medical records from multiple sites and stores them into a single data warehouse. By consolidating patient records in one repository several benefits emerge:

- 1. Enhancing data-cleaning efforts:** centralizing patient records simplifies data-cleaning processing by providing a unified platform for identifying and resolving discrepancies, errors and inconsistencies. This streamlines data management tasks and may ensure that the data used for analysis and reporting are accurate and reliable.
- 2. Identifying and removing duplicate records:** client-level repositories may enable the identification and elimination of duplicate patient records.
- 3. Facilitating data analysis and use:** centralized repositories may improve the ability of programme managers and decision-makers to access and analyse patient-level data for programme monitoring, identify trends and gaps and enhance the use of data for evidence-based decision making and planning.
- 4. Improving patient care and management:** having comprehensive patient records readily available in a centralized repository may improve the access of health-care providers to vital information about patients' medical history, treatment regimens and outcomes. This supports continuity of clinical care for improved patient management.

Box 4 provides an illustrative example from the Western Cape in South Africa on using client-level repositories to improve patient care and data quality.

“Client-level repositories serve as centralized clinical data registries that gather client medical records from multiple sites and stores them into a single data warehouse. By consolidating patient records in one repository several benefits emerge.”

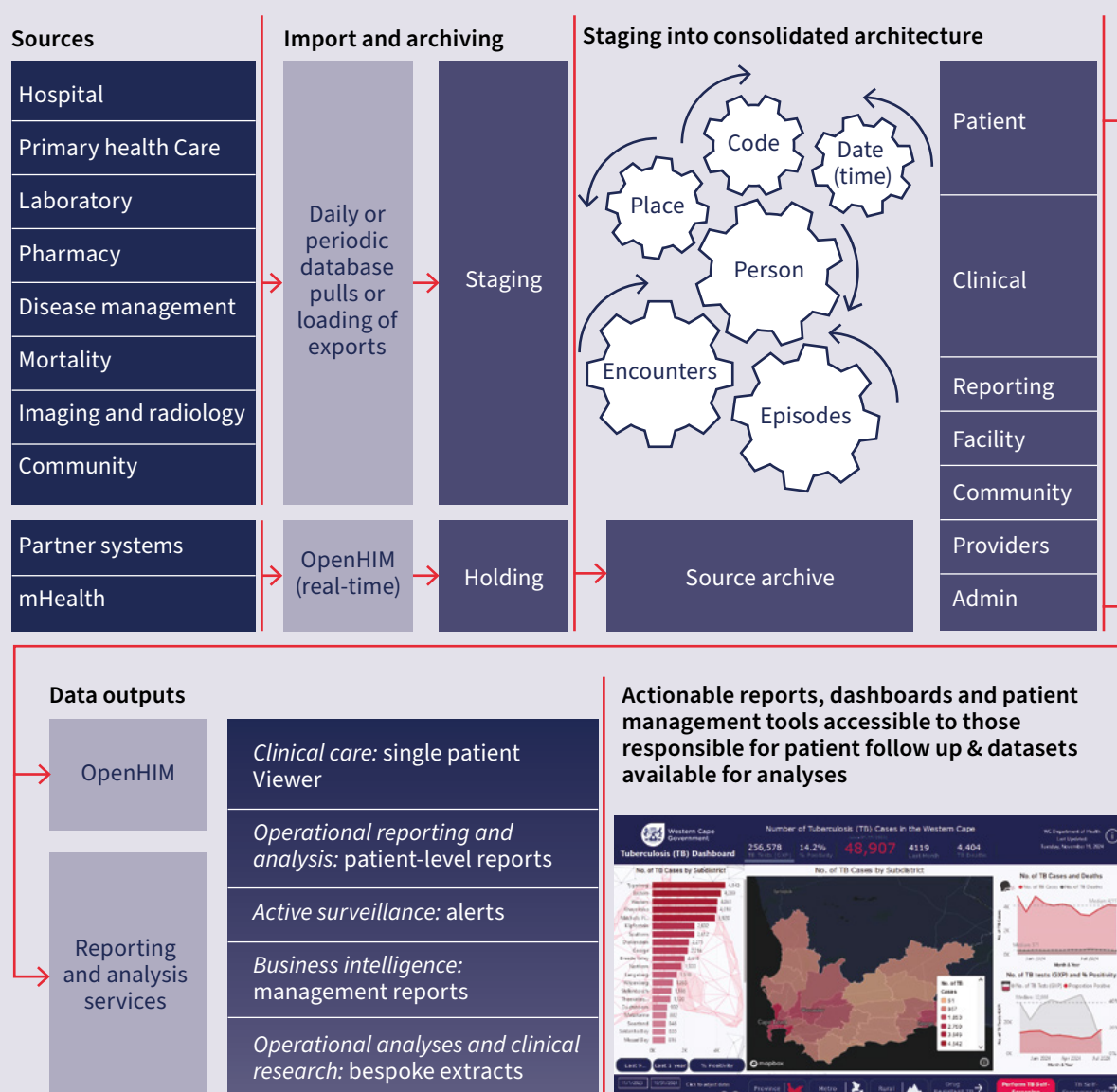


Box 4. Client level repositories in the Western Cape, South Africa

The Provincial Government collaborated with academic and service partners to develop a patient-centred consolidated data environment and suite of digital products to enhance patient care, improve health services delivery and better understand epidemic trends, resources needs and burden in the Western Cape. Data are uploaded from digital public-sector health information systems daily into the Western Cape Provincial Health Data Centre, with storage of the source data, curated clinical repositories and pre-processed episodic cascades or cohorts (Fig. 4).

Fig. 4. The Western Cape Provincial Health Data Centre (PHDC): data harmonization, processing, linkage and tooling processes

High level data architecture



Source: Boulle A, Heekes A, Tiffin N, Mutemaringa T, Zinyakatira N, Phelanyane F, Pienaar C, Buddiga K, Coetzee E, van Rooyen R, Dyers R, Fredericks N, Loff A, Shand L, Moodley M, de Vega I and Vallabhjee K. Data centre profile: the provincial health data centre of the Western Cape Province, South Africa. *Int J Popul Data Sci.* 2019;4(2):1143.

The curated, linked data environment facilitates longitudinal patient records, patient alerts, dashboards, aggregate summary reports, line lists and operational and clinical research. Linked patient records provide a more accurate account of a patient's status since it is a combined record regardless of the number of facilities or hospitals attended (Fig. 5). Clinicians can quickly digest a patient's history without looking through thick paper records. When a patient is marked as lost to follow-up, there is high confidence that the patient has not been seen at any health-care facility for 90 days after the last medication in hand. This makes tracing and recall lists and alerts much more accurate and less overwhelming for the staff actioning the reports and improves data quality. In comparison to facility-drawn tracing lists, those drawn from the Western Cape Provincial Health Data Centre contain, on average, 38% fewer patients due to self-transfers to other facilities and 7% fewer patients due to patients showing up at a hospital without referral (based on a preliminary analysis by staff at selected facilities).

Fig. 5. Patient longitudinal summary



Linked aggregate summaries enable efficient decision-making and advocacy for changes to policy and/or resource allocation based on burden. For example, data showing good viral suppression results for people receiving antiretroviral drugs via repeat prescription collection services (or medication delivered outside the facility pharmacy such as via e-lockers, adherence clubs and community drop-offs) are enabling discussions on longer scripts for people stable on medication. This may ease facility burden, reduce services delivery costs and help to mitigate patient logistics and financial challenges (Fig. 6).

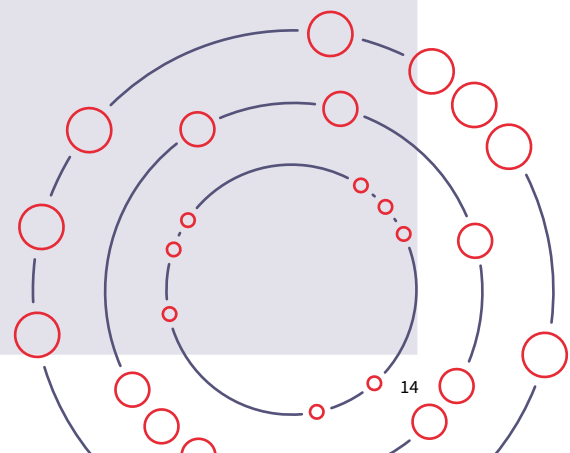


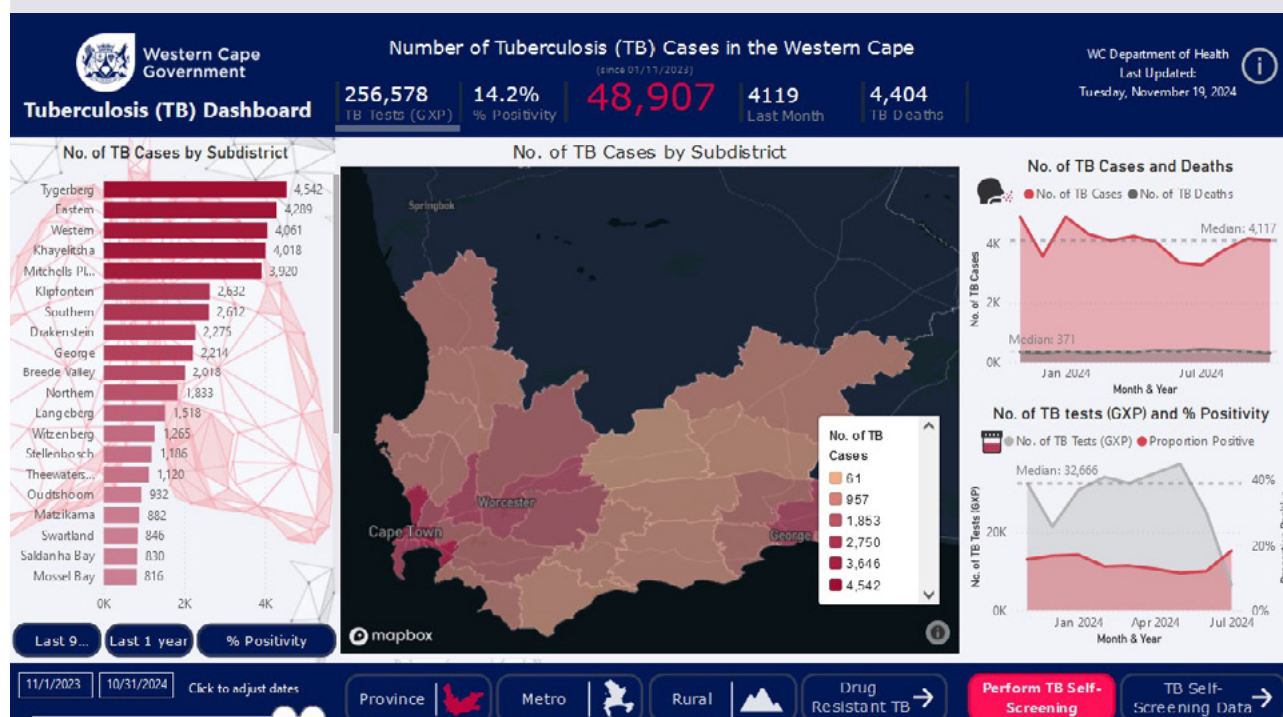
Fig. 6. High viral suppression rates for patients receiving medication from pick-up points outside the facility pharmacy at a metro district facility

To evaluate interventions to improve patient care in an efficient way

- 10 574 people receiving ART
 - **1158** (11%) receiving early ART
 - **9416** (89%) on long-term ART
- Of those on long-term ART
 - **4638** (49%) in repeat prescription collection strategies
- Of those in repeat prescription collection strategies
 - **1769** (37%) in adherence clubs
 - **284** (6%) in facility pick-up point
 - **2674** (57%) unspecified repeat prescription collection strategies
- Of those in repeat prescription collection strategies
 - **32** (<1%) elevated viral load > 1000 copies/mL
 - **167** (4%) suppressed viral load 50–100 copies/mL
 - **4260** (92%) undetectable viral load < 50 copies/mL
 - **179** (4%) viral load not performed > 15 months

Dashboards from the Western Cape Provincial Health Data Centre are either public or internally facing and enable a stakeholder or manager to quickly understand and discuss epidemic trends and burden and acknowledge hot-spots for enhanced focus and service delivery needs (Fig. 7).

Fig 7. Public-facing tuberculosis dashboard



At a time when data breaches are a significant concern, the Western Cape Provincial Health Data Centre (client-level repository) has enabled better governance over data, facilitating deidentified linked data sharing for both operational research (to determine whether interventions become best practices) and for clinical research that informs policy, both positively impacting patient outcomes. Security can also be better coordinated when identified data does not reside outside the protected databases, although the increase in cyberattacks is a growing concern for consolidated data environments housing sensitive personal data.

In summary, health-care delivery can benefit immensely through well-sustained and funded client-level repositories, improving the health outcomes of a population by embracing the patient as a central part of the platform objectives and goals.

Source: Department of Health, South Africa and University of Cape Town, 2024.

4.2.5 Investing in interoperability

WHO defines interoperability as “the ability of different applications to access, exchange, integrate and use data in a coordinated manner, through the use of shared application interfaces and standards, within and across organizational, regional, and national boundaries to provide timely and seamless portability of information that is used to optimize health outcomes” (1).

Interoperability offers numerous benefits, including minimizing errors by reducing manual data entry, ensuring the comprehensiveness of patient records and timely data transfer and availability – all key elements of data quality. Importantly, interoperability enables health-care workers to respond to patient needs more quickly and proactively. Sharing data between facilities also reduces loss to follow-up when individuals migrate to a different location and improves supply chain decision-making by ensuring

that real-time data are available. Strengthening linkage between community information systems and health-care facility systems facilitates patient care, especially with the scale-up of differentiated service delivery implementation. The benefits of interoperability vary depending on how broadly they are adopted – at the point of care, within a facility or between facilities. Regardless, investments in interoperability improve HIV programmes and data quality management amid increasing data volumes and digitalization of data systems.

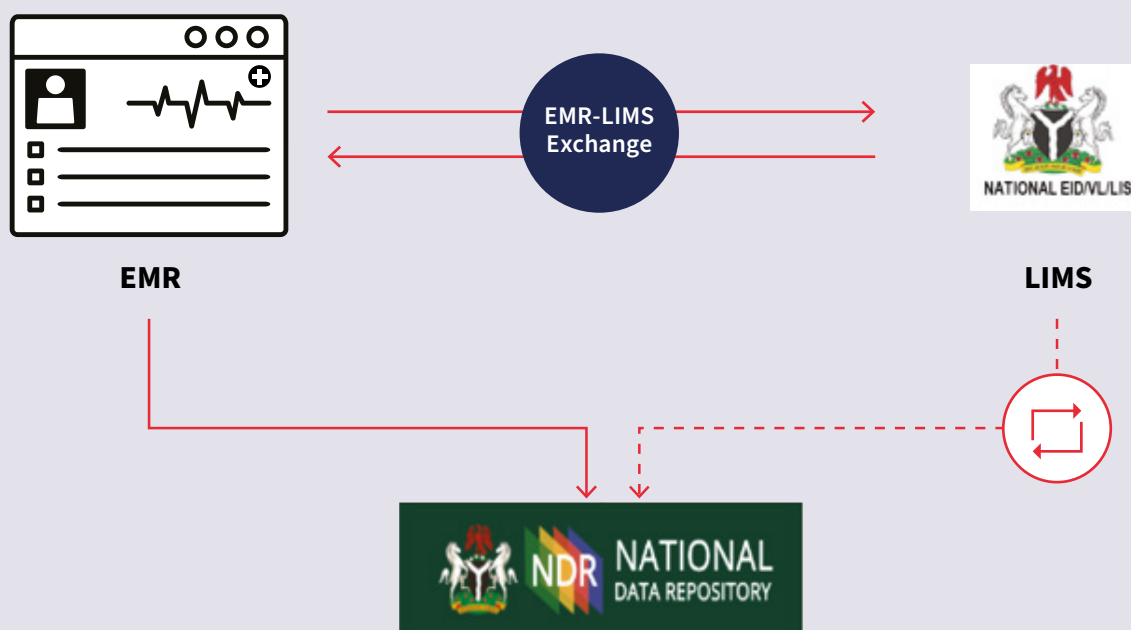
The benefits of interoperability are exemplified in Nigeria’s efforts to enhance its electronic medical record and laboratory information systems. Box 5 highlights Nigeria’s successful implementation of an integrated sample referral network, showcasing the impact of interoperability on patient care and data quality management.

Box 5. Nigeria’s electronic medical record, laboratory information system and national data repository interoperability project

Nigeria supports viral load testing through a network of 17 DNA PCR laboratories across the country in what is called the integrated sample referral network (Fig. 8). The project implemented an electronic data exchange layer between the laboratory information management system and electronic medical records via a structured and secure application programme interface to achieve three key objectives:

- direct logging of sample information from electronic medical records into the laboratory information management system before sample arrival at the reference laboratory;
- prompt retrieval of approved results from the laboratory information management system to electronic medical records electronically; and
- centralized monitoring of PCR laboratories across the country using the National Data Repository.

Fig. 8. Data linkage and flow between the electronic medical records, laboratory information management system and National Data Repository



EMR: electronic medical records; LIMS: laboratory information management system; NDR: National Data Repository.

Strategies deployed for improved monitoring and interoperability between electronic medical records and the laboratory information management system for viral load testing in Nigeria range from harmonization and alignment of concepts and data formats, establishment of regulatory frameworks, integration of existing relevant systems, data security, infrastructure and human capacity building, user-friendly interfaces and continuous evaluation and feedback. In addition, efforts are currently underway to link the laboratory information management system to the National Data Repository for sample management, equipment monitoring (performance, error flags etc.), commodities and consumables stock monitoring, human resource management and optimization.

The system has been widely adopted, with 18 of the 19 states supported by the United States Centers for Disease Control and Prevention already on board and more states in the process of joining. Implementation has significantly reduced transcription errors, enhancing the completeness and accuracy of result documentation. This improvement is critical for patient monitoring and management and for accurate reporting and informed decision-making.

In addition, the system has reduced the turnaround time for sample logging and results retrieval in user facilities. This efficiency ensures that electronic medical records are promptly updated and that reporting to the National Data Repository is conducted without delay. Eliminating manual data entry at PCR laboratories and facilities has lightened the workload, enabling staff to focus more on patient care and other essential tasks.

Source: National AIDS and STI Programme, Federal Ministry of Health, Nigeria, 2024.



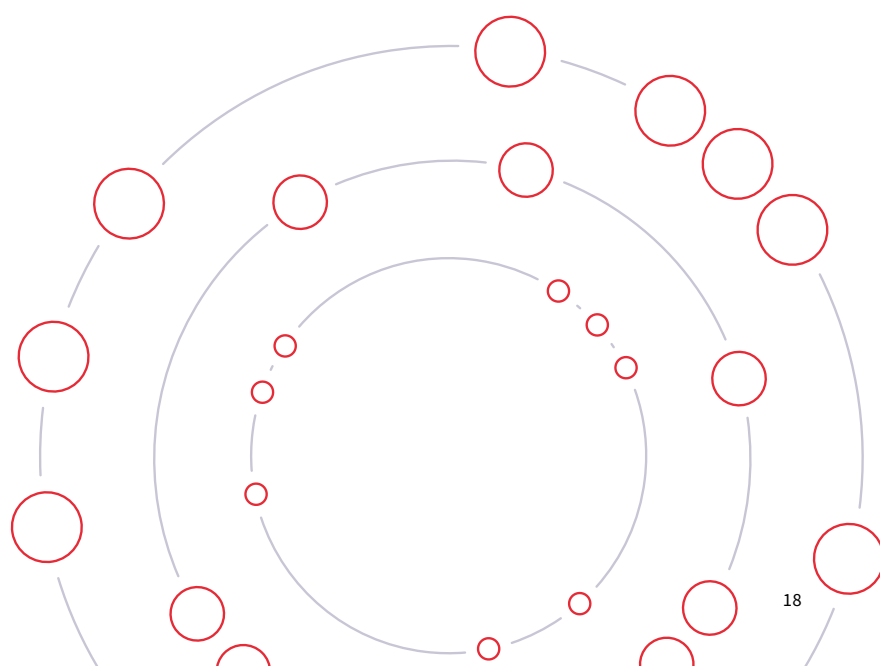
Photo credit ©WHO - Barry Christianson

5. Conclusion

Countries and partners have invested considerable resources and efforts to enhance data quality in accordance with WHO HIV strategic information guidelines for programme improvement. The costs of inaccurate data in country programmes can be huge, affecting programme allocation, budgeting and the assessment of performance. By investing in data quality management, programmes can ensure efficient use of resources, minimize waste, optimize supply chains, and achieve significant cost savings. These cost-effective investments may improve sustainable HIV responses that adapt to evolving needs.

As HIV programmes transition to differentiated service delivery, including community-delivered services, and digital data ecosystems in countries evolve, data quality challenges going forward include the need for integration and linkage between different data systems facilitated by using unique identifiers to ensure the timely flow and availability of data and the safe and secure exchange of growing volumes of health data. Routinizing data quality management approaches and enhancing data utilization may help to sustain the HIV response. This can be supported by using digital solutions, potentially leveraging AI and machine learning with robust data protection measures alongside digital data collection and automated data validation to strengthen data systems.

Moreover, moving beyond solely focusing on DQA and adopting a comprehensive data quality management approach encompassing activities that contribute to long-term DQI strategies strengthen country data systems. This includes investing in interoperability, data deduplication, data visualization, analytics and capacity building to enhance data quality and utilization, ultimately strengthening service delivery and programme impact and ensuring sustainable returns on investments.



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